



ABBOTSLEIGH

FILE

AUGUST 2006

YEAR 12

ASSESSMENT 4

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators may be used
- A table of standard integrals is provided with this paper
- All necessary working should be shown in every question

Total marks - 84

- Attempt Questions 1-7
- All questions are of equal value
- Answer each question in a separate writing booklet

Total marks – 84
Attempt Questions 1–7
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Differentiate $y = \sin^{-1}(x^2)$.

2

(b) Use the substitution $u = 4 - x$ to evaluate $\int_0^3 \frac{x}{\sqrt{4-x}} dx$.

3

(c) Evaluate $\int_1^{\sqrt{2}} \frac{dx}{\sqrt{2-x^2}}$.

2

(d) Find $\int \sin^2 3x dx$.

2

(e) Solve $\frac{1}{x+2} \leq 2$

3

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve the equation $\sin 2\theta = 2\cos^2 \theta$ for $0 \leq \theta \leq 2\pi$.

3

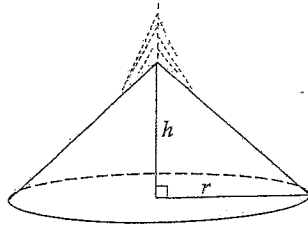
(b) (i) Express $\sin x + \cos x$ in the form $R \sin(x + \alpha)$.

2

(ii) Hence, or otherwise, sketch the graph of $y = \sin x + \cos x$, showing the endpoints, in the domain $0 \leq \theta \leq 2\pi$.

2

(c) Wheat runs from a hole in a silo at a constant rate and forms a conical heap whose base radius, r , is three times its height, h .



NOT TO SCALE

After 1 minute, the height of the heap is 20 cm. Find:

(i) The volume of the conical heap after 1 minute. Leave your answer in terms of π .

2

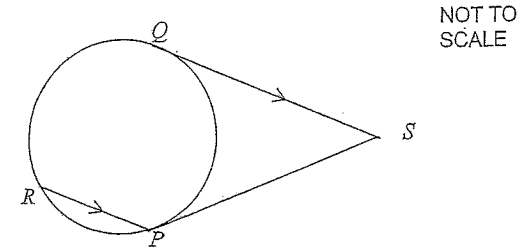
(ii) The rate at which the height of the heap is rising after 1 minute.

3

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



3

P and Q are points on a circle and the tangents to the circle at P and Q meet at S . R is a point on the circle so that RP is parallel to QS .

Copy or trace the diagram into your writing booklet.

Prove that $QP = QR$.

(b) (i) Find the value of k if $2x + 1$ is a factor of $P(x) = 2x^3 - x^2 + kx + 1$.

2

(ii) Show that when k has this value, $P(x)$ has only the one real root, $x = -\frac{1}{2}$.

3

(c) The rate at which a body cools in air is proportional to the difference between its temperature, T and the constant temperature, P of the surrounding air. This rate can be expressed by

$$\frac{dT}{dt} = k(T - P)$$

(i) Show that $T = P + Ae^{kt}$ where A and k are constants, satisfies this equation.

1

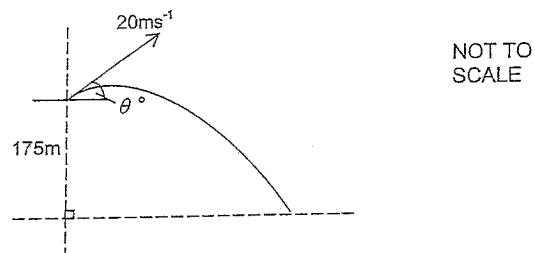
(ii) A heated body cools from 40° to 30° in 1 hour. The air temperature around the body is 20° C. Find the temperature of the body after a further 2 hours.

3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) When $(3+x)^n$ is written as a polynomial in x , the coefficient of x^4 is twice the coefficient of x^3 . Find the value of n .
- (b) A man standing on top of a vertical cliff throws a stone into the air at an angle θ to the horizontal. The top of the vertical cliff is 175 metres above a flat sea.



The angle of projection to the horizontal is θ , as shown.
Let (x, y) be the position of the stone at time t seconds after being thrown.
The initial velocity of the stone is 20 ms^{-1} .

You may assume that the path of the stone where the acceleration due to gravity is -10 ms^{-2} , is given by the parametric equations below

$$x = 20t \cos \theta$$

$$y = 20t \sin \theta - 5t^2 + 175$$

(Do NOT prove these equations.)

The angle of projection of the bullet to the horizontal, θ is 30° .

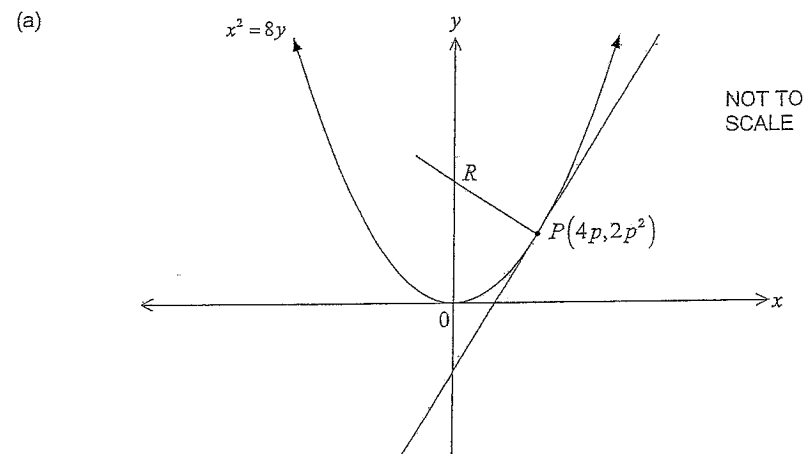
- (i) Find the time it takes for the stone to hit the water.
- (ii) Find the speed at which the stone hits the water.
- (c) A particle moves on a line so that its distance from the origin at time t seconds is x cm. Its acceleration is given by

$$\frac{d^2x}{dt^2} = 10x - 2x^3$$

- (i) If its velocity is v and the particle changes direction 1 cm to the right of the origin, find v^2 in terms of x .
- (ii) Explain why the particle can never reach the origin.

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks



Consider the variable point $P(4p, 2p^2)$ on the parabola $x^2 = 8y$.

- (i) Prove that the equation of the normal at P is $x + py = 2p^3 + 4p$.
- (ii) Find the coordinates of the point R , where the normal at P intersects the y -axis.
- (iii) If M is the midpoint of PR , find the equation of the locus of M in Cartesian form.

- (b) Use mathematical induction to prove that for all integers $n = 1, 2, 3, \dots$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

- (c) The three roots of the polynomial equation $x^3 - 3px^2 + 2qx - r = 0$ form an arithmetic series.

Prove that $r = 2p(q - p^2)$.

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) (i) Sketch the graph of the function $f(x) = x^3 - 1$ in the domain $0 \leq x \leq 2$, clearly showing the coordinates of any points of intersection with the axes. 1

(ii) On the same diagram sketch the graph of the inverse function $f^{-1}(x)$, showing the coordinates of any points of intersection with the axes. 2

(iii) Explain why the x -coordinate of any point of intersection of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $x^3 - x - 1 = 0$. 1

(iv) Show that the equation $x^3 - x - 1 = 0$ has a root between $x = 1$ and $x = 2$. 2

(v) Taking $x = 1.5$ as the first approximation to the root, use one application of Newton's method to find a better approximation to the root, correct to 3 significant figures. 3

(b) The coefficient of x^k in $(1+x)^n$ where n is a positive integer is denoted as $\binom{n}{k}$.

Prove that $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + (n-1)\binom{n}{n-1} = n(2^{n-1} - 1)$ 3

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve the equation $(n + \frac{1}{n})^2 - 5(n + \frac{1}{n}) + 6 = 0$. 2

(b) (i) Show that the function $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ has no stationary points. 3

(ii) Prove that the lines $y = \pm 1$ are asymptotes. 2

(iii) Sketch the curve. 1

(iv) If k is a positive constant, show that the area in the first quadrant enclosed by the above curve and the lines $y = 1$, $x = 0$ and $x = k$ is given by

$$\text{Area} = k - \ln(e^k + e^{-k}) + \ln 2$$

(v) Prove that for all positive values of k , this area is always less than $\log_e 2$. 2

END OF PAPER

QUESTION 1

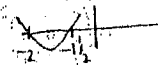
a) $\frac{dy}{dx} = \frac{2x}{\sqrt{1-x^2}}$
 $= \frac{2x}{\sqrt{1-x^2}}$

b) Let $u = 4-x$ $\therefore du = -dx$
 $x=3, u=1$
 $x=0, u=4$

$\therefore I = \int_4^1 \frac{4-u}{\sqrt{u}} x - du$
 $= \int_1^4 \frac{4-u}{\sqrt{u}} du$
 $= \int_1^4 (4u^{-1/2} - u^{1/2}) du$
 $= [2x^4 u^{1/2} - \frac{2}{3} u^{3/2}]_1^4$
 $= 8 \times \sqrt{4} - \frac{2}{3} \times 4 \times \sqrt{4} - (8 - \frac{2}{3})$
 $= 16 - \frac{16}{3} - 8 + \frac{2}{3} = 3\frac{1}{3}$

c) $\int \frac{dx}{\sqrt{1-x^2}}$
 $= [\sin^{-1} \frac{x}{\sqrt{2}}]_{\frac{1}{\sqrt{2}}}^{\frac{\sqrt{2}}{2}}$
 $= \sin^{-1} \frac{\sqrt{2}}{\sqrt{2}} - \sin^{-1} \frac{1}{\sqrt{2}}$
 $= \frac{\pi}{2} - \frac{\pi}{4}$
 $= \frac{\pi}{4}$

d) $I = \int \frac{1}{2} (1 - \cos 6x) dx$
 $= \frac{1}{2} [x - \frac{\sin 6x}{6}] + C$
 $= \frac{x}{2} - \frac{\sin 6x}{12} + C$

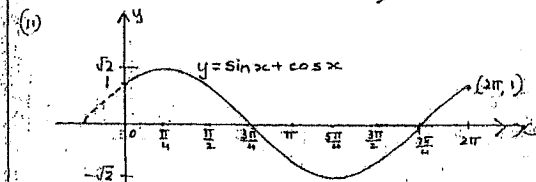
e) $\frac{1}{(x+2)} \leq 2$ $x \neq -2$, x b.s. by $(x+2)^2$
 $(x+2) \leq 2(x+2)^2$
 $(x+2)(2(x+2)-1) \geq 0$
 $(x+2)(2x+3) \geq 0$

 $x < -2$ or $x > -1\frac{1}{2}$

QUESTION 2

a) $\sin 2\theta = 2 \cos^2 \theta$
 $2 \sin \theta \cos \theta = 2 \cos^2 \theta$
 $\cos \theta (\sin \theta - \cos \theta) = 0$
 $\cos \theta = 0$ or $\tan \theta = 1$
 $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{4}$ or $\frac{5\pi}{4}$

b) Let $\sin x + \cos x = R \sin(x+\alpha)$
 (1) $\sin x + \cos x = R \sin x \cos \alpha + R \cos x \sin \alpha$
 $R \cos \alpha = 1$ (1)
 $R \sin \alpha = 1$ (2)
 (2) $\div$ (1) $\tan \alpha = 1$
 $\alpha = \frac{\pi}{4}$
 (1)² + (2)² $R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1+1$
 $R^2 = 2$
 $R = \sqrt{2}$

$\therefore \sin x + \cos x = \sqrt{2} \sin(x + \frac{\pi}{4})$

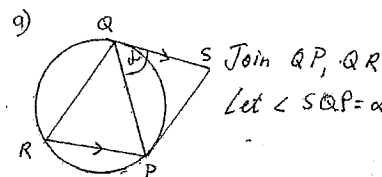


c) $V = \frac{1}{3} \pi r^2 h$ $r = 3h$
 (i) $\therefore V = \frac{1}{3} \pi (3h)^2 \times h$
 $= 3\pi h^3$
 at $t=1, h=20 \therefore V = 3\pi (20)^3$
 $= 24000\pi \text{ cm}^3$

(ii) $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$
 $\frac{dV}{dh} = 9\pi h^2$ at $t=1, h=20$
 $\therefore \frac{dV}{dh} = 9\pi (20)^2$
 $= 3600\pi$
 $\frac{dV}{dt} = 24000\pi$ (from (i))
 $\therefore \frac{dh}{dt} = \frac{1}{3600\pi} \times 24000\pi$
 $= \frac{20}{3}$

$\therefore$ height is rising at $\frac{20}{3}$ cm/minute

QUESTION 3



$\angle QRP = \angle PQS = \alpha$ ($\angle$ between a tangent and a chord = $\angle$ in alt. segment)
 $\angle RPQ = \angle SQP = \alpha$ (alt. $\angle^s$, $RP \parallel QS$)
 $\therefore \angle QRP = \angle QPR = \alpha$
 $\therefore \Delta QRP$ is isosceles
 $\therefore QR = QP$ (sides opp. $= \angle^s$ in isosc. Δ)

b) (i) If $2x+1$ is a factor, $P(-\frac{1}{2}) = 0$
 $2(-\frac{1}{2})^3 - (-\frac{1}{2})^2 + K(-\frac{1}{2}) + 1 = 0$
 $-\frac{1}{4} - \frac{1}{4} - \frac{K}{2} + 1 = 0$
 $\therefore K = 1$

(ii) $\therefore P(x) = (2x+1) \cdot Q(x)$
 $= (2x+1)(x^2-x+1)$
 For x^2-x+1 to have a root, $\Delta \geq 0$
 $\Delta = (-1)^2 - 4 \times 1 \times 1$
 $= -3$
 < 0
 $\therefore x^2-x+1$ has no roots
 So $x = -\frac{1}{2}$ is the only root.

c) (i) If $T = P + Ae^{kt}$
 $\frac{dT}{dt} = Ake^{kt}$
 $= k \times Ae^{kt}$
 $= k \times (T - P)$
 $\therefore T = P + Ae^{kt}$

c) continued

(ii) $P = 20, t=0, T=40$ + $t=1, T=30$
 $T = 20 + Ae^{kt}$
 $40 = 20 + Ae^0 \Rightarrow A = 20$
 $\therefore T = 20 + 20e^{kt}$
 $30 = 20 + 20e^{k}$
 $e^k = \frac{10}{20} = 0.5$
 $k = \ln 0.5$
 $\therefore T = 20 + 20e^{\ln 0.5 \times t}$
 $t = 3, T = 20 + 20e^{\ln 0.5 \times 3}$
 $= 22.5^\circ$
 $\therefore$ temperature is 22.5° after 3 hours

QUESTION 4

1) T_5 has x^4 term
 coeff of $x^4 = {}^nC_4 \cdot 3^{n-4} \times 1^4$
 coeff of $x^3 = {}^nC_3 \cdot 3^{n-3} \times 1^3$
 coeff of $x^4 = 2 \times$ coeff of x^3
 $\therefore \frac{n(n-1)(n-2)(n-3)}{4 \times 3 \times 2 \times 1} \times 3^{n-4}$
 $= \frac{2 \times n(n-1)(n-2)}{3 \times 2 \times 1} \times 3^{n-3}$
 $\therefore \frac{(n-3)}{4} \times 3^{-1} = 2$
 $\frac{(n-3)}{4} \times \frac{1}{3} = 2$
 $n-3 = 24$
 $n = 27$

2) $x = 20t \times \cos 30^\circ$
 $x = 20t \times \frac{\sqrt{3}}{2}$
 $= 10\sqrt{3}t$
 $\dot{x} = 10\sqrt{3}$

$y = 20t \times \sin 30^\circ - 5t^2 + 175$
 $y = 10t - 5t^2 + 175$

(i) stone hits water at $y=0$
 $5t^2 - 10t - 175 = 0$
 $t^2 - 2t - 35 = 0$
 $(t-7)(t+5) = 0$
 $t-7=0$ or $t+5=0$
 $\therefore t=7$ or -5

but $t > 0 \therefore t = 7$
 $\therefore$ It takes 7 seconds to hit the water.

(ii) at $t=7$, $\dot{x} = 10\sqrt{3}$

$\dot{y} = 10 - 10t$
 $\therefore \dot{y} = 10 - 10 \times 7$
 $= -60$

speed $= \sqrt{(10\sqrt{3})^2 + (-60)^2}$
 $= 62.4 \text{ m/s}$



c) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 10x - 2x^3$

(i) $\frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} + C$

$v=0$, $x=1$

$\therefore 0 = 5 - \frac{1}{2} + C$

$\therefore \frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} - 4\frac{1}{2}$

$v^2 = 10x^2 - x^4 - 9$

(ii) For the particle to reach the origin,
 $x=0$

$\therefore v^2 = -9$ which is impossible

$\therefore$ The particle can never reach the origin.

QUESTION 5

a) (i) $x^2 = 8y \Rightarrow y = \frac{x^2}{8}$

$\frac{dy}{dx} = \frac{2x}{8}$
 $= \frac{x}{4}$

at $P(4p, 2p^2)$, m of tangent $= \frac{4p}{4}$
 $= p$

$\therefore$ m of normal is $-\frac{1}{p}$ since

$m_1 m_2 = -1$ for perpendicular lines.

eq'n of normal: $y - 2p^2 = -\frac{1}{p}(x - 4p)$

$py - 2p^3 = -x + 4p$

$\therefore x + py = 2p^3 + 4p$ is the

eq'n of the normal.

(ii) at R , $x=0$

$\therefore 0 + py = 2p^3 + 4p$

$y = 2p^2 + 4$

$\therefore R(0, 2p^2 + 4)$

(iii) $M\left(\frac{4p^3}{2}, \frac{2p^3 + 2p^2 + 4}{2}\right)$

$\therefore m(2p, 2p^2 + 2)$

$x=2p$ into $y = 2p^2 + 2$

$y = 2\left(\frac{x}{2}\right)^2 + 2$

$\therefore$ eq'n of locus is $y = \frac{x^2}{2} + 2$

b) Show that the result is true for $n=1$

LHS $= 1n!$

$= 1$

RHS $= (1+1)! - 1$

$= 2 - 1$

$= 1$

$=$ LHS $\therefore$ result is true for $n=1$

Assume that the result is true for $n=k$
 i.e. assume $1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$

Using this assumption, show that the result is true for $n=k+1$.

i.e. show $1 \times 1! + 2 \times 2! + \dots + k \times k! + (k+1)(k+1)!$

$= (k+2)! - 1$

is true

LHS $= (k+1)! - 1 + (k+1)(k+1)!$

$= (k+1)! (1 + k+1) - 1$

$= (k+1)! (k+2) - 1$

$= (k+2)! - 1$

$=$ RHS

$\therefore$ if the result is true for $n=k$ then it is also true for $n=k+1$

By the process of mathematical induction, this result is true for all integers $n=1, 2, 3, \dots$

c) $P(x) = x^3 - 3px^2 + 2qx - r = 0$

let roots be $\alpha-d, \alpha, \alpha+d$.

sum of roots: $\alpha-d + \alpha + \alpha+d = 3p$

$3\alpha = 3p$

$\alpha = p$ (1)

sum of product of roots, 2 at a time:

$\alpha(\alpha-d) + \alpha(\alpha+d) + (\alpha-d)(\alpha+d) = 2q$

$\alpha^2 - \alpha d + \alpha^2 + \alpha d + \alpha^2 - d^2 = 2q$

$3\alpha^2 - d^2 = 2q$

but $\alpha = p \therefore 3p^2 - d^2 = 2q$

$d^2 = 3p^2 - 2q$

product of roots: $(\alpha-d) \cdot \alpha \cdot (\alpha+d) = r$

$\alpha(\alpha^2 - d^2) = r$

$\alpha^3 - \alpha d^2 = r$

$p^3 - p(3p^2 - 2q) = r$

$\therefore r = p^3 - 3p^3 + 2pq$

$= -2p^3 + 2pq$

$r = 2p(q - p^2)$

or from (1) $\alpha = p$ is a soln so

$P(\alpha) = 0$ and $P(p) = 0$

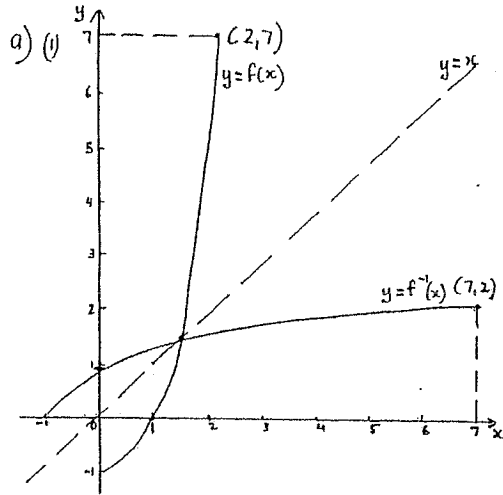
$\therefore p^3 - 3p^3 + 2qp - r = 0$

$-2p^3 + 2qp - r = 0$

$r = 2qp - 2p^3$

$r = 2p(q - p^2)$

QUESTION 6



b) $(1+x)^n = 1 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n$
 Differentiating both sides with respect to x :
 $n(1+x)^{n-1} = {}^nC_1 + 2 {}^nC_2 x + 3 {}^nC_3 x^2 + \dots + (n-1) {}^nC_{n-1} x^{n-2} + n {}^nC_n x^{n-1}$
 Let $x=1$,
 $n \cdot 2^{n-1} = {}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \dots + (n-1) {}^nC_{n-1} + n$
 ${}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \dots + (n-1) {}^nC_{n-1} = n \cdot 2^{n-1} - n$
 $\therefore ({}^nC_1) + 2({}^nC_2) + 3({}^nC_3) + \dots + (n-1)({}^nC_{n-1}) = n(2^{n-1} - 1)$

(iii) Any function and its inverse must intersect in the line $y=x$ since this is their axis of symmetry.
 Pt of intersection of $y=x^3-1$ and $y=x$
 $x^3-1=x \therefore x^3-x-1=0$

(iv) If $x=1$, $g(x) = 1-1-1 = -1 < 0$

If $x=2$, $g(x) = 8-2-1 = 5 > 0$

$g(x)$ changes sign $\therefore$ there is a root between $x=1$ and $x=2$

(v) $g'(x) = 3x^2-1$

Let $x_1 = 1.5$

$x_2 = x_1 - \frac{g(x_1)}{g'(x_1)}$
 $= 1.5 - \frac{(1.5)^3 - 1.5 - 1}{3(1.5)^2 - 1}$

$= 1.3478\dots$

≈ 1.35

$\therefore$ a better approximation to the root is $x=1.35$

QUESTION 7

a) let $m = n + \frac{1}{n}$

$\therefore m^2 - 5m + 6 = 0$

$(m-3)(m-2) = 0$

$m = 3 \text{ or } 2$

$\therefore n + \frac{1}{n} = 3 \text{ or } n + \frac{1}{n} = 2$

$n^2 - 3n + 1 = 0 \text{ or } n^2 - 2n + 1 = 0$

$n = \frac{3 \pm \sqrt{9-4}}{2} \quad (n-1)^2 = 0$

$\therefore n = \frac{3 \pm \sqrt{5}}{2} \text{ or } 1$

b) (i) $\frac{dy}{dx} = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$
 $= \frac{e^{2x} + 2e^{-2x} - e^{2x} + 2e^{-2x}}{(e^x + e^{-x})^2}$
 $= \frac{4}{(e^x + e^{-x})^2}$

$\neq 0$ for any value of x

$\therefore$ The function has no stationary points.

(ii) If $y=1$, then $\frac{e^x - e^{-x}}{e^x + e^{-x}} = 1$

$e^x - e^{-x} = e^x + e^{-x}$

$\therefore 2e^{-x} = 0$

$e^{-x} = 0$ which is

not possible $\therefore y \neq 1$

If $y=-1$ then $\frac{e^x - e^{-x}}{e^x + e^{-x}} = -1$

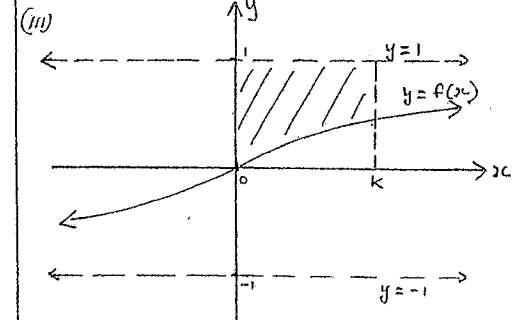
$e^x - e^{-x} = -e^x - e^{-x}$

$2e^x = 0$

$\therefore e^x = 0$ which is

impossible.

$\therefore y = \pm 1$ are asymptotes



(iv) Area = $k - \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$
 $= k - \left[\ln(e^x + e^{-x}) \right]_0^k$
 $= k - \ln(e^k + e^{-k}) + \ln(e^0 + e^0)$
 $= k - \ln(e^k + e^{-k}) + \ln 2$

(v) Area = $k \times \ln e - \ln(e^k + e^{-k}) + \ln 2$
 $= \ln e^k - \ln(e^k + e^{-k}) + \ln 2$
 $= \ln \frac{e^k}{e^k + e^{-k}} + \ln 2$

since e^k and e^{-k} are both > 0 for all k ,
 $\frac{e^k}{e^k + e^{-k}} < 1$ for all k

$\therefore \ln \frac{e^k}{e^k + e^{-k}} < 0$

$\therefore$ Area $< \ln 2$ for all values of k